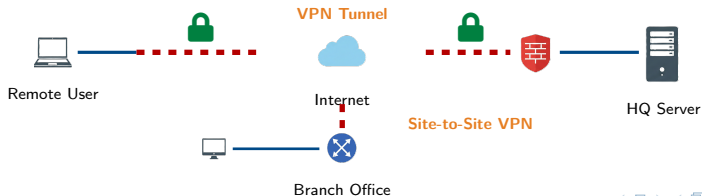


WAN, VPNs, and Remote Management

Intro to Networks – Module 13.0

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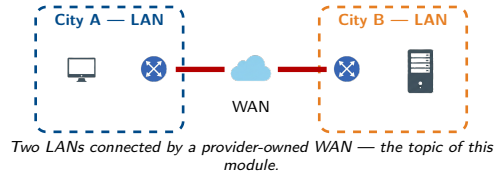
Outline I

- 1 WAN and Internet Connectivity
- 2 Virtual Private Networks
- 3 Remote Management

Review: From Wireless LANs to Wide Area Networks

Key Module 12 Concepts

- **802.11 (Wi-Fi)** connects devices on a local wireless network.
- An **Access Point (AP)** bridges wireless clients to the wired LAN.
- Wi-Fi covers a building or campus — but *not* the distance between cities.



The Question for Module 13

How do networks in *different cities or countries* communicate securely? That requires **WAN links**, **VPNs**, and **remote management** tools.

Learning Outcomes I

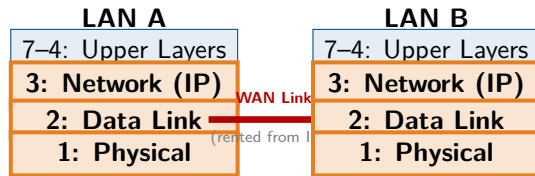
After completing this module, you will be able to:

- Describe common **WAN** technologies and choose the right one for a scenario.
- Distinguish between **Fiber to the Curb (FTTC)** and **Fiber to the Premises (FTTP)**.
- Explain what a **VPN** is and why it is needed for secure remote communication.
- Describe how **tunneling** and **encapsulation** protect data in transit.
- Identify common tunneling protocols: PPTP, L2TP/IPsec, OpenVPN, WireGuard.
- Explain **IPsec** and the difference between **AH** and **ESP**, and Transport vs. Tunnel mode.
- Describe the two phases of **IKE** (Internet Key Exchange).
- Compare **client-to-site**, **clientless**, and **site-to-site** VPNs.
- Use remote management tools: **SSH**, **Telnet**, **RDP**.
- Explain **out-of-band management**, **jump boxes**, and **API** access.

Section 1: WAN and Internet Connectivity

What Is a WAN?

- A **Wide Area Network (WAN)** connects networks that are geographically separated — different buildings, cities, or countries.
- Unlike a LAN, you **do not own** the WAN infrastructure. You *rent* it from an **Internet Service Provider (ISP)**.
- WANs operate across OSI **Layers 1–3**: physical media, data link framing, and IP routing at the edges.



WANs operate at Layers 1–3. Upper-layer processing happens at each LAN endpoint.

Everyday Example

Loading a website: home LAN → ISP WAN → internet → server. You own the LAN; the ISP owns the WAN.

Internet Access Types: Overview

Technology	Medium	Typical Speed	Shared?	Best For
DSL	Copper phone line	1–100 Mbps	No	Suburban, rural
Cable (DOCSIS)	Coaxial cable	25 Mbps–1 Gbps	Yes	Urban, suburban
Fiber (FTTP)	Fiber optic	100 Mbps–10 Gbps	No	Best performance
Fixed Wireless	Radio	25–300 Mbps	Yes	Rural, backup
Satellite (LEO)	RF (low-orbit)	25–300 Mbps	Yes	Remote locations
Cellular (4G/5G)	RF	10 Mbps–1 Gbps	Yes	Mobile, failover

Shared vs. Dedicated

A **shared** medium means your neighbors use the same capacity. Speeds drop when the network is busy — for example, cable internet often slows on weekday evenings.

Key Vocabulary

Bandwidth: maximum possible data rate

Latency: delay in milliseconds per packet

Throughput: actual delivered speed

DSL and Cable Internet

DSL — Digital Subscriber Line

- Uses existing **telephone copper** wiring to the home.
- **Dedicated** line per subscriber — neighbors do not share your speed.
- Speed drops with **distance** from the ISP's central office.
- **ADSL**: asymmetric — faster download than upload.
- Customer device: **DSL modem** (plus router).



Cable — DOCSIS

- Uses **coaxial cable** (same physical cable as TV service).
- **Shared** among homes in a neighborhood node.
- **DOCSIS** (Data Over Cable Service Interface Specification) is the industry standard.
- Customer device: **cable modem** (plus router).

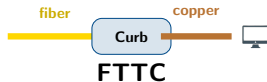


DSL uses dedicated copper to a DSLAM at the central office; cable uses shared coaxial cable to a CMTS. Both backhaul to the ISP via fiber.

Fiber to the Curb vs. Fiber to the Premises

FTTC — Fiber to the Curb / Node

- Fiber reaches a nearby street cabinet.
- The final 100–300 meters use **copper**.
- Cheaper to deploy because fiber stops short of the building.
- The copper tail limits speed, especially as distance grows.



FTTP / FTTH — Fiber to the Premises / Home

- **Pure fiber** runs all the way to the home.
- An **ONT** converts light to Ethernet indoors.
- Delivers the best speed and latency because there is no copper bottleneck.
- Costs more because the ISP must run fiber to each building.



FTTC stops fiber at a street cabinet and uses copper for the final run. FTTP runs fiber directly to an ONT at the home.

Wireless WAN: Cellular and Satellite

Cellular WAN (4G LTE / 5G)

- A **cellular router** with a SIM card provides broadband over the mobile network.
- Common uses: primary WAN at remote sites, **failover backup** if wired WAN fails, IoT devices in the field.
- **5G sub-6 GHz**: broad coverage, 100–500 Mbps typical.
- Bandwidth is **shared** with all mobile users on the tower.

Satellite Internet

- **GEO (Geostationary)**: satellite at 35,786 km. **High latency** ≈ 600 ms RTT — poor for voice/video.
- **LEO (Low-Earth Orbit)**: constellations at ~ 500 km. **Low latency** ≈ 20 –40 ms. Example: Starlink.
- Best for: locations with no wired or cellular option.
- Both are weather-sensitive and have shared capacity.

Real-World Use: Automatic Failover

Many businesses use cellular WAN as a **secondary** / **backup** link. If the primary circuit fails, the router switches to LTE automatically — called **WAN failover**.

Case Study: Tails Sets Up the Freedom Fighter Network

► Case Study: Tails Sets Up the Freedom Fighter Network

Tails must choose WAN connectivity for three outposts, each needing **50 Mbps** minimum.

Outpost	Situation
Alpha — Central City	A fiber ISP has service available at the building.
Beta — Mystic Ruins	No fiber or cable; a 5G tower is visible 2 km away.
Gamma — Space Colony ARK	Only satellite service can reach this location.

Tails also notices: Gamma has a **1.2-second audio delay** on video calls; Beta drops to **30 Mbps** during peak hours.

Review Questions

- 1 Which WAN technology should each outpost use, and why?
- 2 What causes Beta's speed to drop, and what is one solution?
- 3 What causes Gamma's 1.2-second audio delay?

Case Study Solution: Tails' Freedom Fighter Network

✓ Solution: Tails' Freedom Fighter Network

- ❶ **Alpha → FTTP fiber:** Available and provides the best speed and lowest latency.
Beta → Cellular WAN (5G): No wired option; the nearby 5G tower can meet 50 Mbps with good signal. Use a cellular router with a directional antenna pointed at the tower.
Gamma → LEO satellite: Only satellite reaches the ARK. Use LEO (e.g., Starlink), *not* GEO, to keep latency manageable.
- ❷ **Cause:** Cellular is a *shared* medium; tower congestion during peak hours reduces throughput.
Solution: Add a second cellular modem on a different carrier (carrier diversity) or use **WAN bonding** to combine two cellular links.
- ❸ **GEO satellite latency.** GEO satellites orbit at 35,786 km; the round-trip signal propagation alone is ~600 ms, producing a 1.2-second conversation delay. Switching to LEO (≈ 40 ms RTT) would largely eliminate the problem.

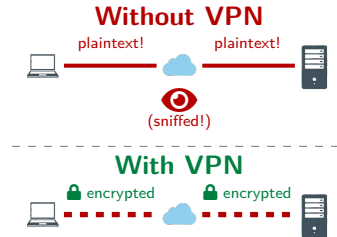
Key Lesson

Match the WAN technology to site constraints: availability, speed requirement, latency tolerance, and budget. There is rarely one correct answer for all sites.

Section 2: Virtual Private Networks (VPNs)

What Is a VPN?

- The public internet is **not private**.
- A **VPN** creates a secure path *through* that untrusted network.
- Every VPN depends on three ideas:
 - 1 **Tunneling** — wraps your packet inside another packet.
 - 2 **Encryption** — scrambles data so only the endpoints can read it.
 - 3 **Authentication** — proves each side is who they claim to be.



Without a VPN, traffic on the internet is readable. A VPN encrypts the data so that only the two endpoints can understand it.

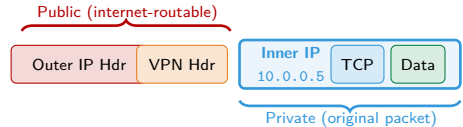
Tunneling and Common VPN Protocols

How Tunneling Works

- **Encapsulation:** the original private packet is wrapped inside a *new*, publicly-addressed outer packet.
- The outer packet travels across the internet normally.
- At the far **tunnel endpoint**, the outer wrapper is removed and the original packet is delivered.
- Think of it as a “**packet within a packet.**”

Tunneling $\neq$ Security

Tunneling alone only *wraps* the packet. You still need **encryption** to keep the contents private!

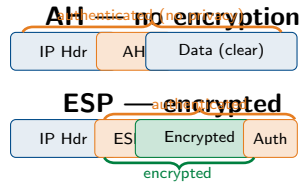


Encapsulation: the private inner packet is wrapped inside a public outer packet for transit across the internet.

Protocol	Encrypted?	Status
PPTP	Weak (broken)	Avoid
L2TP/IPsec	Yes	Common
OpenVPN	Yes (TLS)	Widely used
WireGuard	Yes	Modern, fast
SSTP	Yes (TLS/443)	Windows

IPsec: Security Built Into Layer 3

- **IPsec** (Internet Protocol Security) is a *suite* of protocols that adds security directly to IP at Layer 3.
- **Two operating modes:**
 - **Transport mode:** encrypts only the payload; IP header stays visible. Host-to-host.
 - **Tunnel mode:** encrypts the entire packet, adds new outer IP header. Used in VPN gateways.
- **Two sub-protocols:**
 - **AH:** integrity and authentication — **no encryption**.
 - **ESP:** integrity, authentication, and **encryption**. Standard VPN choice.



AH authenticates the entire packet but does not encrypt. ESP both authenticates and encrypts — always use ESP when privacy is required.

Use ESP, Not AH Alone

AH provides *no confidentiality*. For any VPN carrying private data, always use **ESP**.

Internet Key Exchange (IKE)

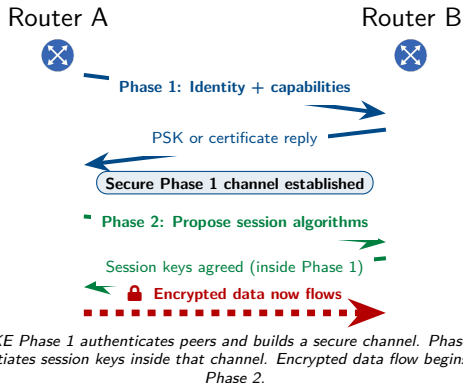
- Before IPsec works, both peers must agree on algorithms and keys.
- **IKE (Internet Key Exchange)** automates that negotiation.
- The result is stored in a **Security Association (SA)**.

Phase 1 — Authenticate

Peers authenticate with a **PSK** or **certificate** and create a secure channel.

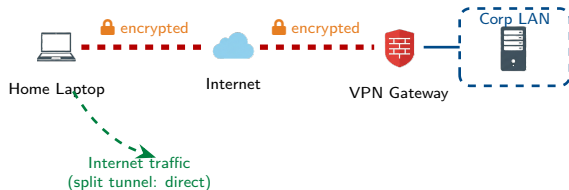
Phase 2 — Negotiate Session Keys

Inside that secure channel, peers choose IPsec settings and generate the session keys for data traffic.



Client-to-Site VPNs

- Also called a **remote access VPN**.
- A user runs **VPN client software** and connects to a corporate gateway over the internet.
- The gateway assigns a **virtual IP address**, so the device behaves like it is on the office network.



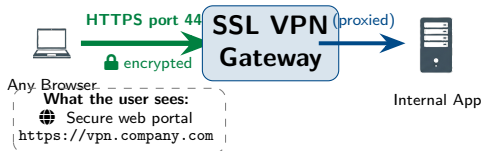
Client-to-site VPN: the user's VPN software creates an encrypted tunnel to the corporate gateway. With split tunnel, internet-bound traffic bypasses the VPN.

Full Tunnel vs. Split Tunnel

- **Full tunnel:** all traffic uses the VPN. More secure, but heavier on corporate bandwidth.
- **Split tunnel:** only company traffic uses the VPN. More efficient, but less isolated.

Clientless VPNs (SSL VPN)

- A **clientless VPN** uses a **web browser** instead of VPN software.
- The user visits a secure portal, logs in, and the gateway **proxies** access to internal web resources.
- Best for contractors, guests, or short-term access from unmanaged devices.



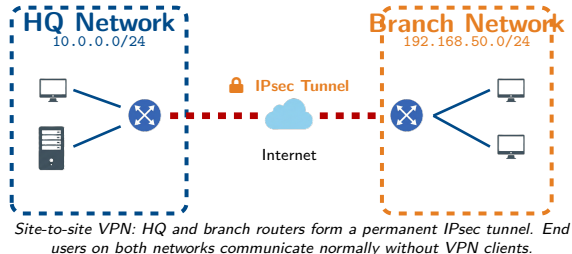
Clientless SSL VPN: the browser connects to a secure web portal; the gateway proxies access to internal web resources. No VPN client software is needed.

Limitation

It mainly works for **web-based resources**; it does not replace a full VPN for RDP or file shares.

Site-to-Site VPNs

- A **site-to-site VPN** connects **two networks**, not two individual users.
- The tunnel is built between **routers or firewalls**.
- End users need no VPN software because the connection is transparent.
- Common use: branch-to-HQ or data-center links.



What Users Experience

Users just connect to remote resources normally; the gateways handle the VPN in the background.

VPN Types: Comparison

VPN Type	Who Uses It	Software on User?	Protocol	Best Use Case
Client-to-Site	Remote workers	Yes (VPN client)	IPsec, TLS	Work-from-home employees
Clientless (SSL)	Contractors, guests	No (browser only)	HTTPS/TLS 443	Web resource access; unmanaged devices
Site-to-Site	Network-to-network	No (on routers)	IPsec	Branch office connectivity

Network+ Exam Tip

Ask three questions: one user, browser-only access, or two full office networks?

Quick Memory Aid

Person + software installed → client-to-site

Browser only, no install → clientless SSL

Office ↔ office, always-on → site-to-site

Case Study: Shadow and G.U.N. Remote Access

► Case Study: Shadow and G.U.N. Remote Access

Shadow must solve three remote-access problems for G.U.N.:

Situation	Details
A	Three field offices need always-on access to HQ file servers through managed routers/firewalls.
B	Agents on personal laptops need full access from untrusted networks.
C	A contractor needs browser-only access to one internal web portal with no install.

Review Questions

- 1 Which VPN type solves Situation A? What device runs the tunnel?
- 2 Which VPN type is correct for Situation B? Full or split tunnel?
- 3 Which VPN type fits Situation C, and why?

Case Study Solution: Shadow and G.U.N.

✓ Solution: Shadow and G.U.N. Remote Access

- ➊ **Site-to-site VPN:** permanent IPsec tunnel between the HQ firewall and each office router.
- ➋ **Client-to-site VPN:** use a **full tunnel** because the laptops are on untrusted networks.
- ➌ **Clientless SSL VPN:** browser-only access to one web app with no software install.

Key Lesson

Match the VPN type to who is connecting and how much access they need.

Section 3: Remote Management

Remote Management: Overview

- **Remote management** means configuring, monitoring, and troubleshooting network devices **without being physically present**.
- Network gear lives in locked data centers, equipment closets, and remote branch offices. You cannot always walk up to a device.

Tool	Port	Encrypted?	Primary Use
SSH	TCP 22	Yes	Secure CLI access to any device
Telnet	TCP 23	No	Legacy CLI (plaintext — avoid!)
RDP	TCP 3389	Yes	Full GUI desktop (Windows)
Console / OOB	Physical / serial	N/A	Last resort / out-of-band access
Jump Box	Via SSH / RDP	Yes	Secure, audited pivot point
API (REST / NETCONF)	HTTPS 443	Yes	Automation and scripting at scale

Rule of Thumb

If it is not encrypted, it should not be used in production. **Telnet** is the primary offender: replace it with **SSH** everywhere.

Secure Shell (SSH)

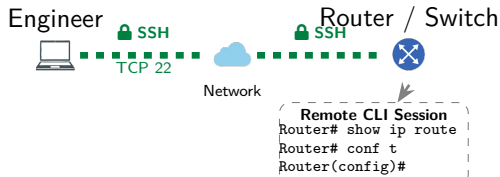
- **SSH** is the standard tool for secure remote **CLI access**.
- It runs on **TCP 22** and encrypts credentials and commands.
- Always use **SSH version 2**.

Authentication Options

- **Password authentication:** simple, but weaker against brute force.
- **Key-based authentication:** stronger because the private key never leaves your device.

Common Commands

```
ssh admin@192.168.1.1
```



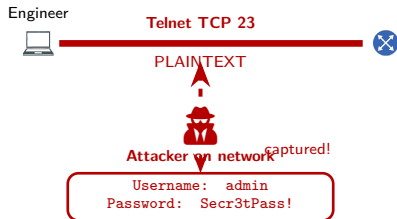
SSH creates an encrypted tunnel (TCP 22) to the device's CLI. All commands and output are protected from eavesdropping along the path.

Telnet: A Cautionary Tale

- **Telnet** was the original remote CLI protocol.
- It uses **TCP 23** and sends everything in **plaintext**.
- Any attacker on the path can capture usernames and passwords.

Never Use Telnet in Production

On real networks, disable Telnet and use SSH instead.



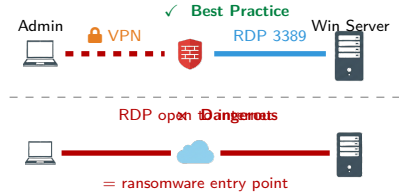
Telnet sends credentials as plaintext. Any attacker with network access can capture usernames and passwords using a packet sniffer.

Remote Desktop Protocol (RDP)

- **RDP** provides a full **graphical desktop session** to a remote Windows system.
- It uses **TCP 3389** and encrypts traffic with TLS.
- Common uses:
 - Managing Windows servers.
 - Help-desk support or remote work access.

Critical Security Warning

Never expose RDP directly to the internet. Put it behind a VPN, firewall, or jump box.



Best practice: reach RDP only through a VPN tunnel. Never expose TCP 3389 directly to the internet.

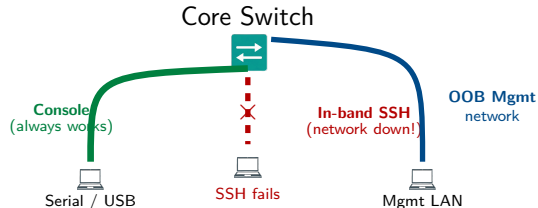
Console Connections and Out-of-Band Management

Console Port Access

- A **console port** is a physical serial or USB management connection.
- It gives CLI access that **bypasses the network entirely**.
- Use it when IP settings are broken or remote access fails.

Out-of-Band (OOB) Management

- **OOB management** uses a **separate management network**.
- If the production network fails, management access can still work.



When the production network is down, the console port (direct physical connection) and OOB management network provide alternative access paths to the device.

Sysadmin Saying

"When SSH won't connect, the console port is your parachute."

Jump Boxes and API-Based Management

Jump Box (Bastion Host)

- A **jump box** is a hardened server used as the single admin entry point.
- Admins connect to it first, then move to internal devices.
- It provides one audited entry path and keeps internal devices off the public edge.

API-Based Management

- Modern devices expose **APIs** for automated management.
- **REST API**: HTTP plus JSON.
- **NETCONF**: XML over SSH with YANG models.
- **gRPC / gNMI**: common for streaming telemetry.

Case Study: Dr. Eggman's Remote Management Crisis

► Case Study: Dr. Eggman's Remote Management Crisis

Orbot reports four urgent remote-management problems:

Problem	Details
1	Factory switches are unreachable over SSH because the production network is down.
2	A Windows server needs a GUI session for a critical patch.
3	200+ devices need automated daily configuration backups.
4	An untrusted contractor needs browser-only access to one internal web portal.

Review Questions

- 1 What should Eggman use when SSH is unavailable?
- 2 Which tool fits the Windows server, automated backups, and contractor portal?

Case Study Solution: Dr. Eggman's Remote Management

✓ Solution: Dr. Eggman's Remote Management Crisis

- 1 **Console port or OOB management:** use a path that does not depend on the production network.
- 2 **RDP:** use it through a VPN or jump box, never directly from the internet.
- 3 **REST API or NETCONF:** automate backups instead of doing manual SSH logins.
- 4 **Clientless SSL VPN:** browser-only access scoped to one portal.

Module 13 Summary I

WAN and Internet Connectivity:

- A **WAN** connects geographically separated networks over provider-owned infrastructure (Layers 1–3).
- **DSL**: dedicated copper phone line; speed degrades with distance. **Cable (DOCSIS)**: shared coaxial cable; faster but contended at neighborhood node.
- **FTTC**: fiber runs to a street cabinet; copper covers the last 100–300 m. **FTTP/FTTH**: pure fiber to the premises via an ONT; best speed and latency.
- **Cellular WAN** (4G/5G): mobile broadband; used for primary or failover WAN. **Satellite**: GEO = high latency (~600 ms); LEO = lower latency (~40 ms).

Module 13 Summary II

Virtual Private Networks:

- A **VPN** = tunneling + encryption + authentication over an untrusted network.
- **Tunneling / encapsulation**: private packet is wrapped inside a public-addressed outer packet; decapsulated at the far endpoint.
- Protocols: PPTP (broken, avoid); L2TP/IPsec; OpenVPN; WireGuard; SSTP.
- **IPsec**: **AH** (auth/integrity, no encryption) vs. **ESP** (auth/integrity + encryption — use this).
Transport mode (host-to-host); **Tunnel mode** (gateway VPNs).
- **IKE**: Phase 1 authenticates peers and builds a secure channel; Phase 2 negotiates session keys inside that channel.
- **Client-to-site**: individual user + VPN client software; full or split tunnel option.
- **Clientless SSL VPN**: browser only; web resources only; no software needed.
- **Site-to-site**: two routers/firewalls form a permanent tunnel; transparent to end users.

Module 13 Summary III

Remote Management:

- **SSH** (TCP 22): encrypted CLI; key-based auth preferred; always use version 2.
- **Telnet** (TCP 23): plaintext — disable everywhere; replace with SSH.
- **RDP** (TCP 3389): encrypted GUI for Windows; never expose directly to the internet — use VPN or a jump box.
- **Console port**: physical serial/USB connection; bypasses the network entirely; last resort when network access is down.
- **Out-of-band (OOB) management**: dedicated management network independent of the production network; survives production outages.
- **Jump box (bastion host)**: hardened pivot point; admins connect here first, then to internal devices; enables audit logging and limits internal exposure.
- **REST API / NETCONF / gRPC**: programmatic management for automation at scale; REST returns JSON over HTTPS; NETCONF uses XML/YANG over SSH.